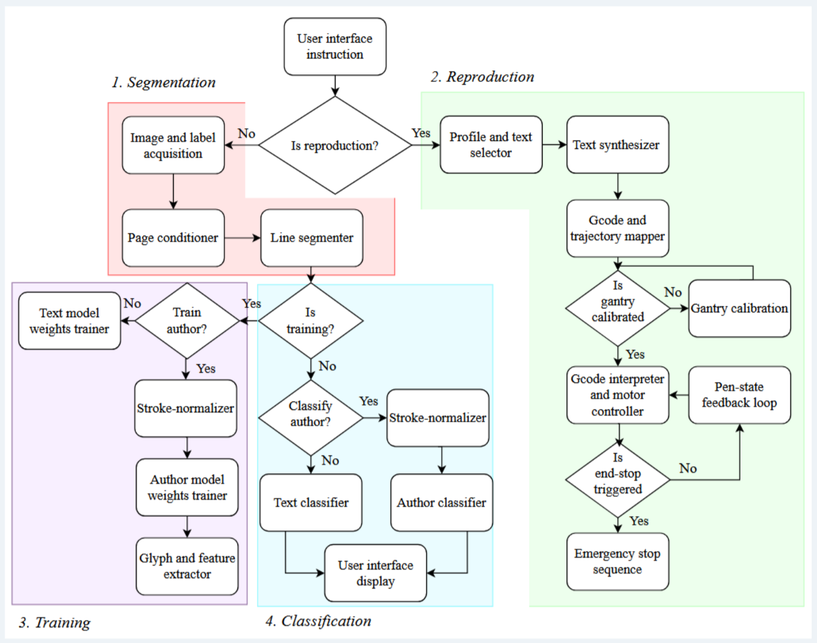
# Handwriting robot

Reading a page and reproducing an individual writing style

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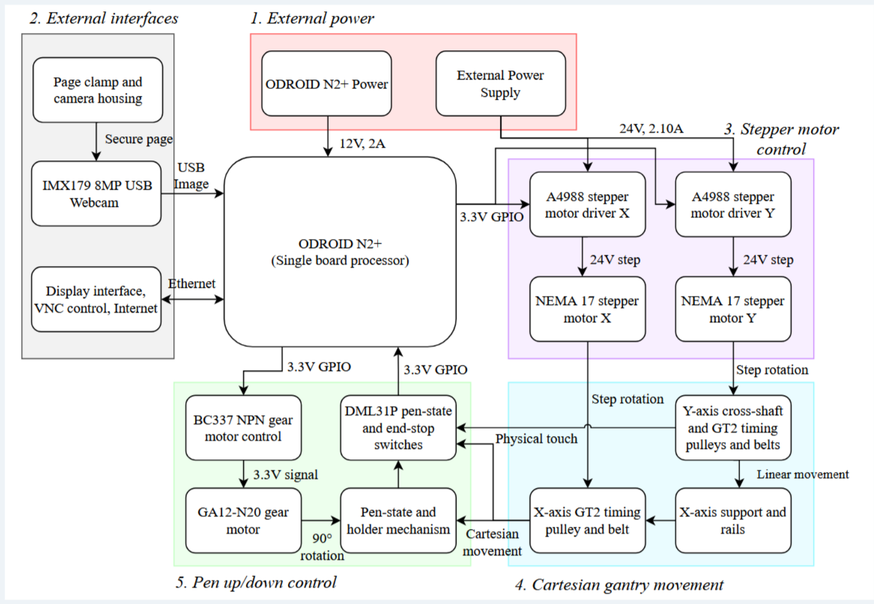



## Project overview

Read photographed handwriting and identify its writer.

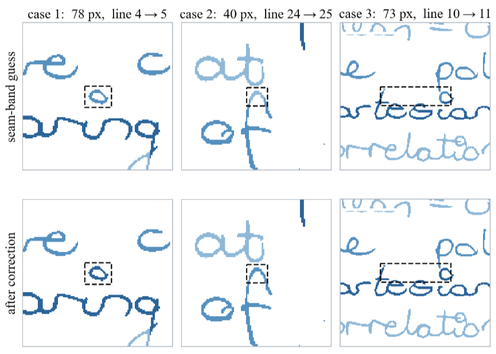
Use stored glyph shapes to write new text in the selected style with a pen gantry.

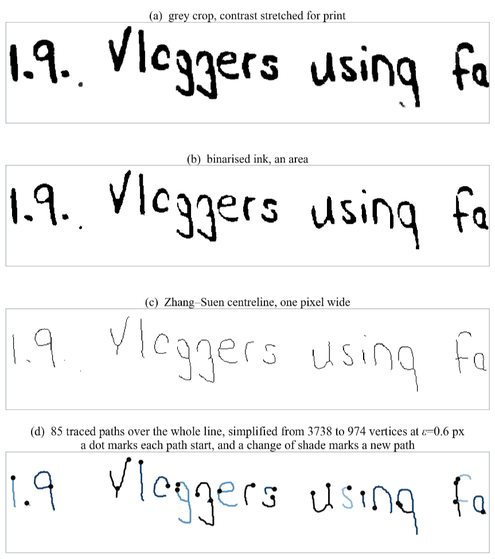
11 writer styles are supported.




## Image processing

Correct lighting and rotation, separate the lines and recover stroke paths from the ink mask.

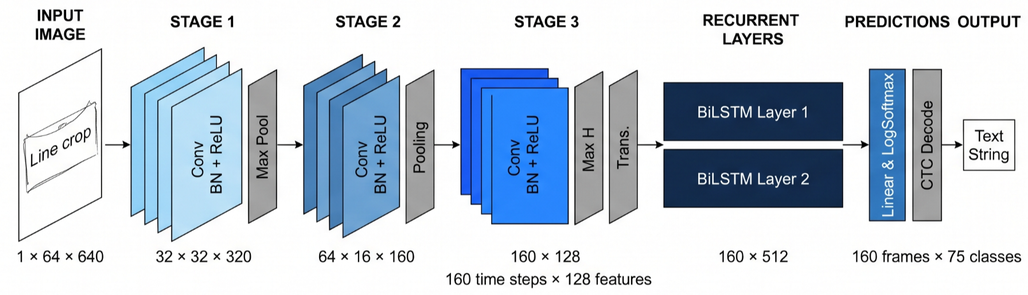



## Recognition

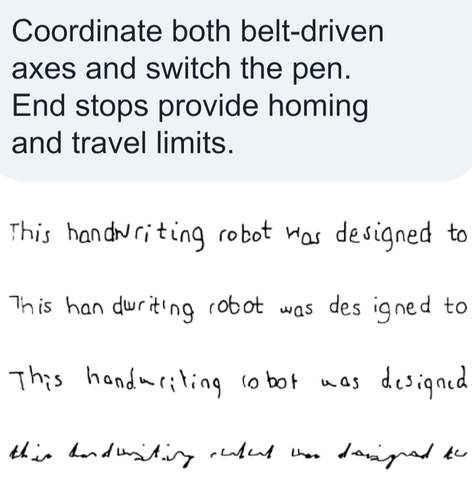
Read the line text with a CNN–BiLSTM–CTC network. Identify the writer using normalised stroke width.

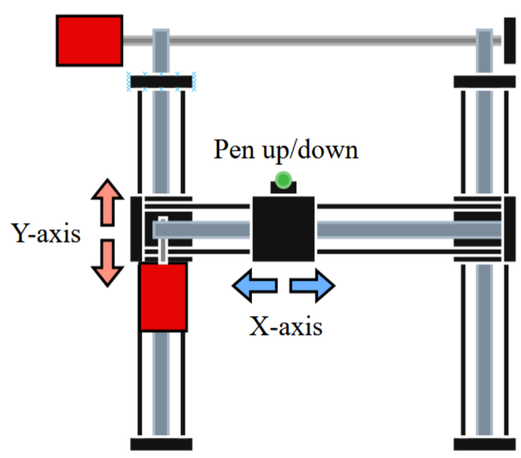
| Parameter | Setting |
|---|---|
| Reader input | 64 × 640 px |
| Writer styles | 11 |
| Stroke width | 0.17 x-height |




## Motion control

Coordinate both belt-driven axes and switch the pen. End stops provide homing and travel limits.

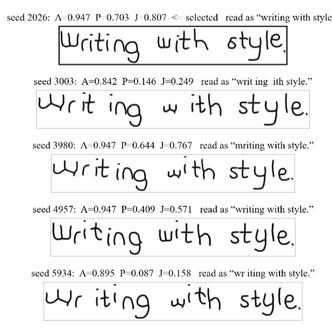


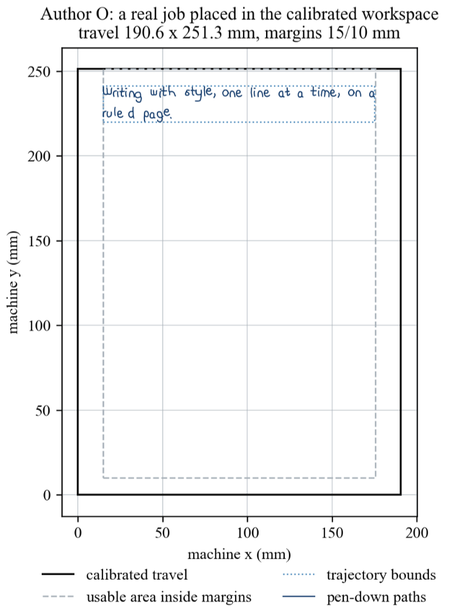




## Handwriting synthesis

Select stored glyphs, set the spacing and joins, then map the paths into G-code.



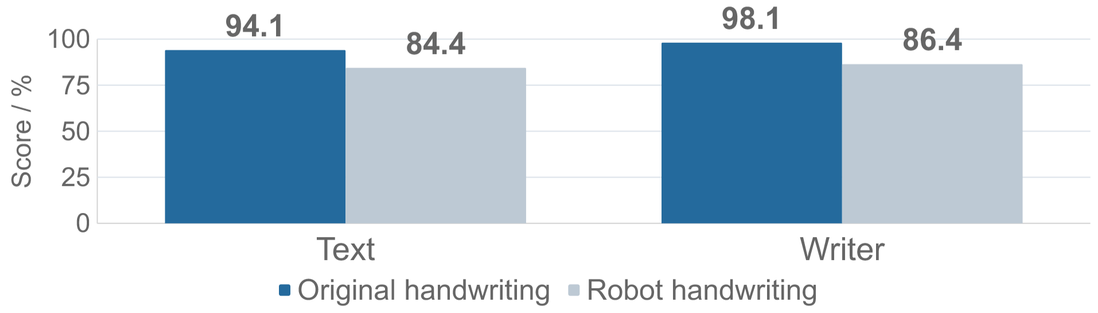

## Measured performance

**86.7%**
**relative shape score**
against natural handwriting variation

**0.49 mm**
largest measured length error

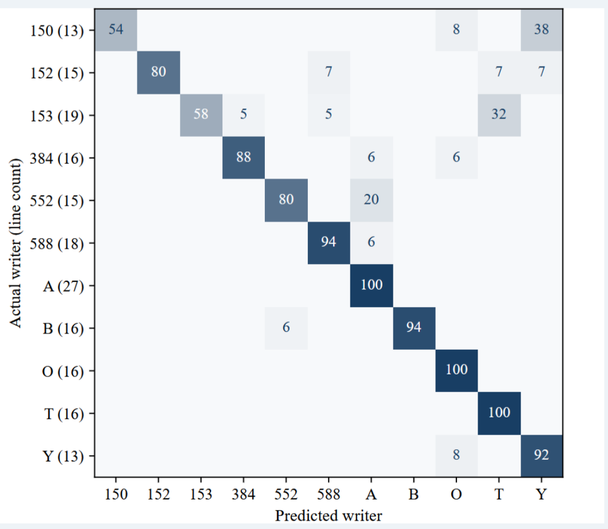
**25 / 25**
completed jobs within 4 seconds per character

## Recognition scores




Original validation and robot output are separate test sets.
Classification time remains above target.

## Robot writer identification




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Handwriting robot

EPR400 project, 2026

11 writer styles, image processing, recognition and physical reproduction